Recovering Motion-Referring Expressions in Moving-Camera Videos via Global Motion Compensation

Anonymous Author(s)

ABSTRACT

Referring multi-object tracking takes a video and a natural-language expression and tracks every object the expression describes. Many expressions refer to how an object moves rather than how it looks, but common hosts read motion from raw box displacement. A moving camera corrupts that signal: a parked car sweeps across the frame and is scored as "moving," while a car keeping pace with traffic can read as static and be dropped. Motion-language RMOT fuses dynamics with text yet leaves the camera in the signal; ego-motion compensation removes the camera but is used only for association. We connect the two with a decision-level plug-in: per-frame homographies compose into a cumulative transform, residual velocity is read at multiple gaps, a two-tower encoder aligns it with the expression, and the score is added to the host logit without host retraining. On iKUN, ego-compensated alignment lifts moving-class HOTA by +8.83 (native $20.25 \rightarrow 29.08$, n=5), and ablation identifies ego-compensation as the only engineered component with a significant effect (-1.77, $p<0.01$). Across three host–split settings, the recovered signal is monotone-inverse to native motion ability. In pooled HOTA the module matches iKUN's published score and improves FlexHook's V2 score outright (+0.281), while FlexHook V1 remains below its published number under a known reproduction gap.

KEYWORDS

referring multi-object tracking, ego-motion compensation, motion-language alignment, plug-in module, Refer-KITTI, video grounding

1 INTRODUCTION

Referring multi-object tracking (RMOT) outputs trajectories for the objects described by a free-form sentence, and many such sentences describe motion rather than appearance. Queries such as "moving cars" or "a vehicle turning left" require actual object motion. However, the usual image-plane displacement mixes two sources: the object's motion and the camera's ego-motion. That mixture fails systematically under a moving camera. A stationary car can sweep across the image and be scored as "moving", while a genuinely moving car can keep a nearly fixed image column when its motion cancels the camera's. The signal a motion-class query depends on is exactly the signal a moving camera corrupts.

The two research lines that could fix this have not met. Motion-language RMOT, including VMRMOT [9] and its RMOT lineage [2, 6, 14], aligns dynamics with text but derives those dynamics from camera-contaminated image-plane trajectories. Plain MOT has camera compensation: BoT-SORT [1] and UCMCTrack [15] estimate camera motion and stabilize predictions, but the corrected geometry is consumed by association and never exposed to language. The missing capability is therefore not merely "GMC plus text". It is a persistent, ego-stable object-motion representation, accumulated over the temporal span of a motion expression, and explicitly aligned with language. Platform pose is in principle available from KITTI GPS/IMU or ground-plane projection, but no RMOT method routes it to language; a feature-based estimate keeps the module sensor-free and host-agnostic.

GMC-Link occupies that intersection. Per-frame homographies estimate camera motion; their cumulative transform gives an ego-stable reference for each track; multi-gap residual velocity turns that reference into a temporal motion descriptor; a contrastive aligner scores whether the descriptor matches the expression; and a calibrated decision-level correction adds only that motion evidence to the host logit. The cumulative transform is central: it makes the descriptor reflect the object rather than the camera and lets the evidence persist across the window a motion verb spans. This yields a falsifiable hypothesis. If ego-compensation matters beyond merely adding motion features, replacing residual velocity with raw velocity should reduce moving-class recovery. Section 4 tests exactly that.

This work connects the mature ego-motion-compensation machinery of plain tracking with the emerging motion–language line of RMOT, and makes the following contributions.

- A plug-in motion-language module that compensates ego-motion, keeps the host detector, tracker, and appearance grounding unchanged, and attaches through a calibrated decision-level correction rather than host retraining.
- A cross-setting characterization over iKUN (V1) and FlexHook (V1/V2) showing that gain follows a monotone-inverse ordering with native motion ability; FlexHook V1/V2 are one architecture on two splits, so they are not independent architectural confirmations.
- Mechanism evidence showing +8.83 moving-class recovery, ego-compensation as the only statistically significant engineered component (-1.77, Welch p=0.006), and decision-level fusion as the viable injection site after feature-level injection regresses.

On iKUN [2], the module matches published pooled HOTA [8] while lifting the motion class hidden by the appearance-heavy pooled metric; on FlexHook [6] V2, it improves pooled HOTA outright. The gains hold at a fixed tracker and are orthogonal to detector quality: reaching the top of the leaderboard requires a stronger tracker than the public detections expose, which is out of scope for a plug-in that changes neither the detector nor the host.

2 RELATED WORK

2.1 Referring MOT and motion-language fusion

TransRMOT [14] established RMOT on Refer-KITTI, iKUN [2] decoupled grounding from association through a frozen tracker, and FlexHook [6] strengthened the two-stage cross-modal hook.

Later RMOT methods improve semantic grounding [4, 5, 7, 10]; VMRMOT [9] even makes motion explicit with position, direction, distance-trend, and speed-trend descriptors. The common limitation is where the dynamics come from: raw image-plane boxes. Thus motion-language RMOT can be language-aware while its motion representation remains entangled with camera ego-motion. TempRMOT [16] already aggregates trajectory history in recurrent memory, so native temporal-memory hosts are out of scope.

2.2 Camera and ego-motion compensation in tracking

Camera compensation is standard in plain MOT, where it stabilizes predictions before association. BoT-SORT [1] estimates an inter-frame transform from ORB matches [12] with RANSAC [3], while UCMCTrack [15] uses a ground-plane camera-motion model. Both improve association under ego-motion, but the corrected geometry is consumed immediately by frame-to-frame association. It is not composed into a persistent ego-stable trajectory, accumulated over the temporal span of a motion expression, or exposed as a language-aligned feature.

2.3 The unfilled intersection

The gap is specific: motion-language RMOT is language-aware but camera-contaminated; camera-compensated MOT is ego-stable but language-unaware. GMC-Link connects them by producing persistent ego-stable motion, aligning it with the expression, and fusing it after the host score. The method therefore preserves host appearance grounding while supplying the motion evidence that the host never had.

3 METHOD

GMC-Link sits between a two-stage RMOT host and its scoring step. The host remains responsible for detection, association, and appearance-language grounding. The GMC branch estimates background camera motion and converts host tracks into ego-stable residual object motion. The aligner asks whether that residual motion matches the referring expression, and fusion adds only a calibrated correction after the host has formed its appearance-based score. This modularity enables post-hoc attachment without replacing the detector, retraining host weights, or hiding the source of a correction. Figure 1 gives the attachment point, and Figure 2 shows the four stages.

The separation is also diagnostic: failures can be assigned to host appearance grounding, ego-motion estimation, motion-language alignment, or scalar calibration instead of to an opaque fused representation.

3.1 Ego-motion compensation

The first stage estimates camera motion between adjacent frames. We use ORB as a sparse feature-point sampler because global camera motion should be estimated from repeatable, matchable background evidence rather than from every image location. Dense sampling or dense optical flow supplies many more measurements, but in driving footage many of those measurements come from weak-texture road regions, object boundaries, reflections, residual parallax, and independently moving foreground objects; these ambiguous samples can introduce high-volume noise and bias a single global transform. ORB instead keeps a compact set of discriminative corners with binary descriptors, making descriptor matching and later outlier rejection more reliable and efficient. We detect ORB keypoints [12], match descriptors with a brute-force Hamming matcher, and keep matches passing Lowe's 0.7 ratio test. Host boxes seed a foreground mask: without masking, correspondences on moving objects can bias the transform away from background camera motion, so keypoints inside tracked objects are suppressed and the surviving matches better represent static scene structure. RANSAC [3] fits $H_{t-1\to t}$ and records median inlier reprojection error as background residual. If too few reliable inliers survive, the stage falls back to identity, so a weak-texture frame degrades to "no compensation" rather than to a corrupted camera-motion estimate.

A single homography is exact only for planar or pure-rotation scenes, but driving footage is sufficiently planar-dominated for this approximation to be useful. In our data, ORB with RANSAC outperformed dense optical flow because explicit feature selection and outlier rejection discard noisy or non-background measurements that a dense field tends to smear into the estimate. Frame-to-frame compensation is still insufficient for language, because motion expressions span multiple frames. We therefore store the original, never-warped centroids and compose the frame-to-frame transforms into one cumulative homography,

$$H_{t-k\to t} = H_{t-1\to t}\, H_{t-2\to t-1} \cdots H_{t-k\to t-k+1}, \tag{1}$$

then apply it once to map a past centroid into the current frame. This creates a persistent ego-stable reference while avoiding the error that would accumulate from repeatedly warping already-warped coordinates.

3.2 Multi-scale residual velocity

For temporal gap g, raw velocity is centroid displacement and contains both object motion and camera-induced displacement. Ego velocity estimates the displacement caused by the camera alone by warping the past centroid through Eq. 1. Residual velocity is their difference,

$$v_g^{\text{res}} = v_g^{\text{raw}} - v_g^{\text{ego}}, \tag{2}$$

which approximates the motion a stationary camera would observe. Thus a parked car has near-zero residual motion even as its pixels move significantly under ego-motion. We normalise by image dimensions, $v_{\text{norm}} = (v_{\text{pix}}/d_{\text{img}}) \times 100$.

We compute residuals at gaps $\{2, 5, 10\}$ and concatenate them with box state into a 13D vector ordered as (dx_s, dy_s), (dx_m, dy_m), (dx_l, dy_l), $\Delta w, \Delta h, c_x, c_y, w, h, snr$; the first six entries are residual (dx, dy) pairs at gaps 2, 5, and 10. The scales are complementary rather than redundant. Gap 2 preserves short-term changes and jitter that long windows average away; gap 10 accumulates slow sustained motion that can be buried in one short step; gap 5 bridges the regimes. This matters for language because expressions may refer to instantaneous changes or sustained movement over different temporal spans. The final *snr* is the mid-scale residual object speed relative to the RANSAC-inlier background noise floor. It does not

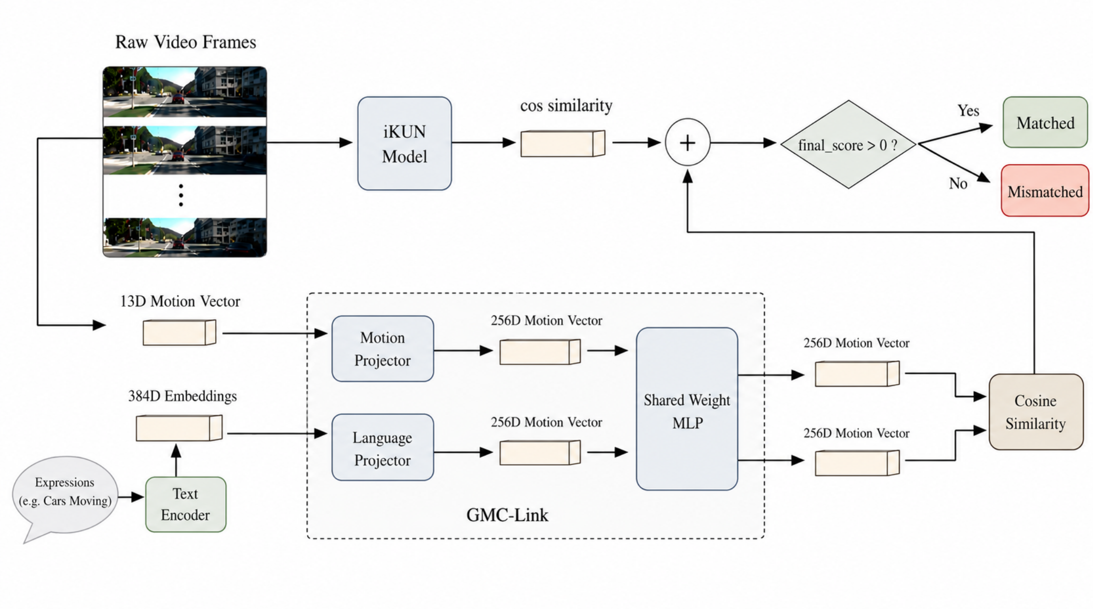

Figure 1: GMC-Link architecture. The host provides tracks and expression scores; GMC-Link estimates camera motion, derives residual motion, aligns it with language, and returns only a calibrated decision-layer correction.

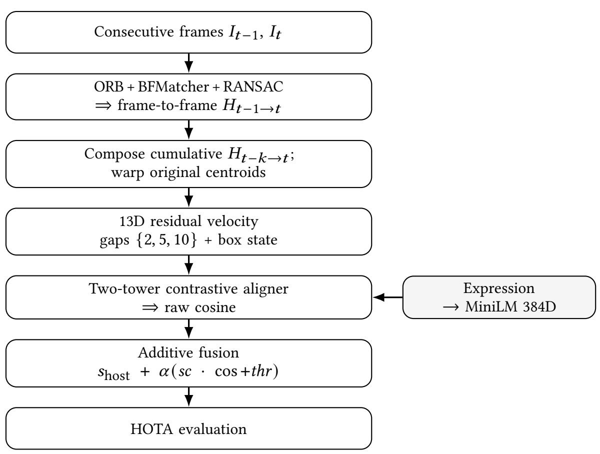

Figure 2: GMC-Link pipeline. ORB homographies compose into a cumulative transform, produce a 13D residual-velocity descriptor, align with the expression, and correct the host logit for HOTA evaluation.

create class separation itself, but represents homography reliability and reduces frame-level score variance when compensation is weak.

3.3 Motion-language alignment

A two-tower encoder maps the 13D motion vector and expression into a shared space. Linear adapters (motion 13 → 256, language 384 → 256 from all-MiniLM-L6-v2 [11]) feed a shared MLP (256 → 512 → 512 → 256), layer norm, and ℓ_2 normalisation. Training uses symmetric InfoNCE [13] at temperature 0.07 with shared-expression false negatives masked. Inference uses raw cosine similarity, with no sigmoid or temporal smoothing, so fusion receives an uncalibrated alignment score. The aligner is intentionally standard: the scientific claim is not a new text encoder or contrastive loss, but that ego-compensated residual motion is the right signal to align. CLIP-text and larger 768D sentence encoders did not consistently improve the shipped raw-cosine pipeline, so MiniLM stays.

3.4 Decision-level additive fusion

The alignment score corrects the host logit at the decision level. We inject it additively per host and per axis,

$$s_{\text{final}} = s_{\text{model}} + \alpha\,\big(sc \cdot \cos + thr\big), \tag{3}$$

where cos is raw motion-language cosine, sc rescales it to the host logit range, thr shifts the operating point, and α sets weight. We use decision-level fusion because feature-level motion injection into the visual encoder regressed an early pipeline by −21.7% F1. The final design therefore preserves the host's appearance representation and injects motion only after the host has formed its score. This keeps appearance grounding intact, prevents motion features from distorting the visual representation, and leaves host evidence and motion evidence as separate auditable terms.

A keyword router selects the motion recipe when an expression contains one of 15 fixed motion substrings (e.g. *moving*, *turning*, *parked*, *braking*); otherwise it selects the appearance recipe. Because motion evidence is largely uninformative for appearance-only expressions, sc_{appear} is seven to eleven times smaller than

Table 1: Locked fusion coefficients for Eq. 3. Appearance scale is damped because motion is uninformative on appearance expressions.

Host	Axis	α	*sc*	*thr*
iKUN	motion	1.00	0.90	+0.17
	appearance	1.00	0.30	+0.10
FlexHook (V1)	motion	0.65	10.0	+3.0
	appearance	1.00	3.50	+0.9
FlexHook (V2)	motion	0.40	10.0	+1.3
	appearance	1.00	3.50	+1.2

sc_{motion}. The coefficients absorb host-specific logit-scale differences, so the table should be read as calibration to each host's score range rather than as a learned fusion head.

Standard-deviation matching and a learned fusion network were tested but underperformed this fixed scalar calibration. This is consistent with the plug-in goal: a new host needs calibration to its score range, but not retraining of its detector, tracker, visual encoder, or language grounding pathway.

4 EXPERIMENTS

4.1 Setup

We evaluate on Refer-KITTI V1 [14] and V2 [16] using TrackEval HOTA [8]. Moving-class HOTA tests whether the module fixes the target failure; pooled HOTA checks whether that recovery survives the full benchmark. Comparing iKUN [2] with FlexHook [6] tests whether the gain concentrates where native motion reasoning is weakest, and the raw-velocity ablation tests whether ego-compensation matters beyond adding any motion descriptor. FlexHook (V1/V2) is one architecture calibrated per split, not two independent methods. Results are mean±std over n=3 seeds for cross-setting tables and n=5 for ablations with Welch tests. The aligner uses AdamW (10^{-3}, batch 256, 100 epochs); ORB uses 1500 features and RANSAC a 5 px threshold. We report native hosts and a simple uncalibrated raw-cosine GMC control to separate adding the signal from calibrating it to the host logit scale.

Refer-KITTI has no held-out tuning split, so Table 1's shipped coefficients are calibrated on the benchmark sequences and should not be claimed as universally transferable hyperparameters. Because these coefficients are benchmark calibration rather than transferable hyperparameters, we use leave-one-sequence-out calibration to bound overfitting risk. iKUN LOSO moving HOTA remains 28.38 ± 1.60 versus 29.47 ± 0.88 in-sample (native 20.25), while FlexHook LOSO pooled is 53.28 ± 0.17 (V1) and 42.74 ± 0.07 (V2), with V2 still above the published 42.526. We therefore claim the iKUN moving-class recovery and the FlexHook V2 pooled gain, not the iKUN pooled number, as calibration-robust.

The test splits are heavily skewed toward appearance. V1 holds 158 expressions over 1010 frames, of which 75.3% are appearance queries, 7.6% are static-motion queries, and only 17.1% (27 expressions) refer to a moving object; V2 holds 862 expressions over 1469 frames with a similar 74.1 / 11.7 / 14.2 split. The motion class our module targets is therefore a roughly one-in-six minority. Pooled HOTA is computed once over the union of all trajectories, so it is dominated by detection and association counts from the appearance majority; it is not a class-weighted mean of per-class HOTA. A recovery concentrated on moving expressions can therefore register as a small pooled movement even when the per-class effect is large. We treat moving-class HOTA as the primary diagnostic and report pooled HOTA as the whole-benchmark check.

Table 2: Pooled HOTA on Refer-KITTI test (n=3, mean±std). Δ is against published pooled HOTA; iKUN matches, while FlexHook V2 is the clean pooled improvement.

Host	Native	+GMC simple	+GMC (ship)	Δ paper
iKUN [2] (V1)	44.564	44.272	44.634 ± 0.066	+0.070
FlexHook (V1) [6]	53.824	53.121	53.526 ± 0.087	−0.298†
FlexHook (V2) [6]	42.526	42.532	42.807 ± 0.038	+0.281

†FlexHook (V1) paper gap is structural (no tested configuration beats 53.824); Δ vs *+GMC simple* anchor = +0.405.

Table 3: Motion deficit across host–split settings. Native moving-class HOTA versus the moving-class HOTA gain added by the module. The gain is monotone-inverse to native motion ability across the three settings.

Host	Native MOVING	GMC MOVING gain
iKUN [2]	20.25	+8.83 ± 0.83 (n=5)
FlexHook (V1) [6]	43.98	+0.91 ± 0.79 (n=3)
FlexHook (V2) [6]	48.02	+0.43 ± 0.17 (n=3)

4.2 Main results

Table 2 first shows that raw GMC evidence is not sufficient by itself: the uncalibrated *+GMC simple* control is negative on iKUN and FlexHook V1. The signal must be calibrated to the host's score range. With the shipped calibration, iKUN [2] reaches 44.634 ± 0.066, matching rather than materially beating the published 44.564, while FlexHook V2 gives the clean pooled improvement, 42.807 ± 0.038 versus 42.526 (+0.281). FlexHook V1 improves over *+GMC simple* (53.121 → 53.526) but remains below its published 53.824.[1] The scientific claim is therefore not that GMC-Link beats every pooled leaderboard number; it is that ego-compensated motion recovers motion-class errors, especially where native motion reasoning is weakest.

4.3 The motion deficit

The gain tracks native motion deficit. iKUN starts lowest and gains most (+8.83 ± 0.83); FlexHook V1 and V2 start higher and gain little (+0.91 ± 0.79 and +0.43 ± 0.17). Across the three settings, the relation is monotone-inverse: larger native deficit receives larger correction. This supports the intended use case, a host that already grounds appearance but does not reliably distinguish ego-induced displacement from object motion. We treat the pattern as a prediction over three settings, not a proportional law.

[1] No tested configuration beats FlexHook V1's published 53.824; the strongest legacy variant reaches 53.716 ± 0.068 (n=3), so we report the gain against *+GMC simple*.

Table 4: Moving-class HOTA ablation on iKUN (n=5). Ego-compensation is the only significant engineered component (−1.77, Welch p=0.006).

Variant	MOVING HOTA
Native host (no module)	20.25
Full module	**29.08 ± 0.83**
−ego (raw velocity)	27.31 ± 0.66
−multiscale (single gap)	28.82 ± 0.84
−snr	29.74 ± 0.59

4.4 Ablations

The ablation directly tests the Introduction hypothesis. If the gain came merely from adding a motion descriptor, raw velocity should perform similarly to ego-compensated residual velocity. Instead, replacing residual velocity with raw velocity costs −1.77 moving-class HOTA (Welch t=3.75, p=0.006). Thus ego-motion removal contributes beyond motion features alone. The full module lifts 20.25 → 29.08 (+8.83); single-gap motion (−0.26, p=0.64) and removing snr (+0.67, p=0.19) stay within seed spread, so ego-compensation is the only statistically significant engineered component here.

Feature-level injection regressed an early host by −21.7% F1, supporting the decision to correct logits after appearance scoring rather than reshaping host representations.

Figure 3 makes the failure mode concrete. The parked car has large raw image displacement because the camera passes it, while the genuinely moving car maintains nearly the same image column. Raw image-plane motion therefore gives the wrong semantic cue in both directions; ego-compensated residual motion restores the distinction. The module runs at 68 FPS (median 14.8 ms/frame excluding host inference), dominated by CPU ORB homography (14.55 ms); the aligner is under 0.2 ms.[2]

5 CONCLUSION AND LIMITATIONS

A decision-level ego-motion module recovers motion-class referring expressions that camera-naive hosts miss. The result suggests that motion-language grounding in moving-camera video should reason over ego-stable trajectories, not raw image-plane displacement. The module keeps the tracker and detector fixed, retrains no host weights, and adds only host-specific scalar calibration. On iKUN [2] it matches published pooled HOTA while lifting the hidden motion class; on FlexHook [6] V2 it improves pooled HOTA outright.

The limits are bounded. A single ORB/RANSAC homography [1] is approximate under non-planar scenes and strong parallax; our evidence is therefore confined to planar-dominated Refer-KITTI driving footage, with no claim of transfer to handheld, aerial, or strongly non-planar video. Calibration is host-specific and benchmark-level coefficients are not universal hyperparameters. We study two host architectures across three host–split settings, with FlexHook V1/V2 not independent architectural confirmations.

[2]An orthogonal appearance re-ranker reaches 45.612 pooled HOTA on iKUN (n=3); we do not claim it here.

Figure 3: Real recovery under camera ego-motion on "moving cars" (Refer-KITTI seq 0005, frames 85/94/100 of a forward-driving clip; the green car moves, the red car is parked). The dashed line marks the mover's image column. **Red:** the parked car's centroid slides left as the camera passes it (center-x 261 ⟶ 104 px), so the camera-naive host scores it "moving" (false positive), yet its ego-compensated residual is near zero and the module restores it to static. **Green:** the moving car holds its column (center-x ≈605, <1 px/frame), so the host misses it (false negative), but its nonzero residual — unlike the static background — recovers it as moving. Both errors flip from the same host logit through the additive correction.

Hosts with native temporal memory, such as TempRMOT [16], remain out of scope.

The broader implication is that camera compensation can serve not only tracking association, but also language grounding. Richer ego-stable trajectory descriptors are the natural next step for camera-naive RMOT hosts.

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